Atom Count Card: Mole Conversions for Bear Brand Sterilized Milk (200 mL)

From a printed label to moles, formula units and atoms of calcium carbonate, $CaCO_3$

CHEMISTRY FOR ENGINEERS
ASYNCHRONOUS ACTIVITY 02

SECTION A · PRODUCT IDENTITY



Compound: calcium carbonate, $CaCO_3$
PubChem CID 10112
Product: Bear Brand Sterilized Milk, 200 mL

Calcium Carbonate ($CaCO_3$) is added as an inorganic stabilizing agent and mineral fortifier to supply "Tibay Support" Calcium ions in UHT sterilized liquid milk frameworks.

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Course: Chemistry for Engineers
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SECTION B · BASE QUANTITATIVE DATA

Parameter	Value	Unit	Traceable source	Precision (s.f.)
Calcium per serving	**112.5**	mg	Product label (primary)	**4 s.f. (1.125×10^2), limiting**
Reference basis	Per 200 mL serving size	mL	Product label	Exact basis
Net packaging contents	200 mL volume basis	mL	Product label	Nominal, not limiting
Chemical formula	$CaCO_3$	—	PubChem CID 10112	Exact subscripts (1, 1, 3)
Atomic weights	C 12.011 · O 15.999 · Ca 40.078	g/mol	IUPAC/CIAAW abridged table (2024)	Not limiting
Avogadro constant	$6.022\,140\,76 \times 10^{23}$	mol^{-1}	BIPM (2019), SI definition	Exact

Assumptions: all calcium originates from $CaCO_3$; calcium is taken as 112.5 mg per 200 mL serving (value supplied by the student); "112.5" has 4 significant figures.

SECTION C · MOLAR-MASS BUILD-UP

Element	Subscript	Atomic wt (g/mol)	Mass contrib. (g/mol)	%E
Ca	1	40.078	40.078	40.044
C	1	12.011	12.011	12.001
O	3	15.999	47.997	47.956
$CaCO_3$	**5 atoms**		**100.086**	**100.000**

Addition rule: M = 40.078 + 12.011 + 3(15.999) = **100.086 g/mol**. Every term has 3 decimals, so M keeps 3 decimals.

Σ%E is exactly 100.000 % before rounding; the rounded entries shown add to 100.001 %.

SECTION D · THE MOLE MAP (200 mL basis)

2.809×10^{-1} g
mass of $CaCO_3$

→ × (1 mol / 100.086 g) →

2.807×10^{-3} mol
moles of $CaCO_3$

→ × (6.022×10^{23}) units/mol →

1.690×10^{21}
formula units of $CaCO_3$

Always pass through moles: grams never convert straight to particles. Name the entity at every step: $CaCO_3$ is ionic, so it is counted in formula units, not molecules.

SECTION E · DIMENSIONAL ANALYSIS PROOFS (unrounded value in gray; reported answer in red)

1 Molar mass

M = 40.078 + 12.011 + 3(15.999) = 40.078 + 12.011 + 47.997 = **100.086 g/mol**

2 Label bridge (all Ca from $CaCO_3$)

$$112.5\ \text{mg Ca} \times \frac{1\ \text{g}}{1000\ \text{mg}} \times \frac{100.086\ \text{g CaCO}_3}{40.078\ \text{g Ca}} = 0.280\,94\ \text{g} \rightarrow \mathbf{2.809 \times 10^{-1}\ g\ CaCO_3}$$

3 Mass → moles

$$0.280\,94\ \text{g} \times \frac{1\ \text{mol}}{100.086\ \text{g}} = 2.8070 \times 10^{-3} \rightarrow \mathbf{2.807 \times 10^{-3}\ mol\ CaCO_3}$$

4 Moles → formula units

$$2.8070 \times 10^{-3}\ \text{mol} \times \frac{6.022\,140\,76 \times 10^{23}\ \text{units}}{1\ \text{mol}} = 1.6904 \times 10^{21} \rightarrow \mathbf{1.690 \times 10^{21}\ formula\ units}$$

5 Oxygen atoms (3:1 exact ratio)

$$1.6904 \times 10^{21}\ \text{units} \times \frac{3\ \text{O atoms}}{1\ \text{unit}} = 5.0713 \times 10^{21} \rightarrow \mathbf{5.071 \times 10^{21}\ O\ atoms}$$

6 Two-route check (calcium) %Ca = [40.078 / 100.086] × 100 % = 40.044 %

Route A: 280.94 mg × 0.40044 = 112.50 mg

$$\text{Route B: } 2.8070 \times 10^{-3}\ \text{mol} \times \frac{1\ \text{mol Ca}}{1\ \text{mol}} \times \frac{40.078\ \text{g}}{1\ \text{mol}} \times \frac{1000\ \text{mg}}{1\ \text{g}} = 112.50\ \text{mg}$$

Both routes → **1.125×10^2 mg Ca ✔ agree**

7 Reverse: % → formula (per 100 g)

Ca 40.044 ÷ 40.078 = 0.99915 · C 12.001 ÷ 12.011 = 0.99917 · O 47.956 ÷ 15.999 = 2.9975

÷ 0.99915 → Ca 1.000 : C 1.000 : O 3.000 → **$CaCO_3$** (empirical = molecular, n = 1)

8 Scale-up, 200 cans

$$200\ \text{cans} \times \frac{2.8070 \times 10^{-3}\ \text{mol}}{1\ \text{can}} = 0.56141\ \text{mol} \rightarrow \mathbf{5.614 \times 10^{-1}\ mol\ CaCO_3}\ (5.619 \times 10^1\ \text{g})$$

SECTION F · MACRO SCENARIO SCALE-UP

1 · ONE 200 mL PACK

1.690×10^{21}

formula units of $CaCO_3$ in one can (2.807×10^{-3} mol)

2 · EXAM WEEK, 200 CANS

3.381×10^{23}

formula units; 5.614×10^{-1} mol, about 5.619×10^1 g $CaCO_3$

3 · DEPARTMENT, ONE YEAR

6.170×10^{25}

formula units; 1.025×10^2 mol, about 1.025×10^4 g. Assumes 100 cans/day × 365 days (counts exact).

4 · THE ONE-MOLE BOUNDARY

3.56×10^2 cans

$$1.00\ \text{mol} \times \frac{100.086\ \text{g}}{1\ \text{mol}} \times \frac{1\ \text{can}}{0.280\,94\ \text{g}} = 356.25\ \text{cans}$$ (3 s.f., set by 1.00 mol)

SECTION G · SIGNIFICANT FIGURES AUDIT

Calculator: 1.690 43 × 10^{21} units → **1.690×10^{21} formula units**

Final precision is constrained to 4 significant figures due to the 112.5 mg calcium limitation printed on the manufacturer's 200 mL nutritional panel. Exact: N_A, subscripts, can counts, 1 g = 1000 mg. The one-mole result keeps 3 s.f. because 1.00 mol has 3. Guard digits were kept until the last step.

ONE-NUMBER TAKEAWAY

One 200 mL can holds about 1.690×10^{21} $CaCO_3$ formula units, yet about 3.56×10^2 cans are needed to collect a single mole: a mole is an enormous count, but only 100.086 g of matter.

SECTION H · REFERENCES (APA 7TH EDITION) & AI USE DISCLOSURE

Bureau International des Poids et Mesures. (2019). *The International System of Units (SI)* (9th ed.). https://www.bipm.org/en/publications/si-brochure
Commission on Isotopic Abundances and Atomic Weights. (2024). *Abridged standard atomic weights*. International Union of Pure and Applied Chemistry. https://www.ciaaw.org/abridged-atomic-weights.htm
National Center for Biotechnology Information. (n.d.). *PubChem compound summary for CID 10112, calcium carbonate*. Retrieved [date accessed], from https://pubchem.ncbi.nlm.nih.gov/compound/Calcium-carbonate
[Manufacturer name]. (n.d.). *Bear Brand Sterilized Milk, 200 mL* [Product label]. Photograph by D. B. P. Gacer, [Month Day, Year].
AI use disclosure: AI was used to enhance the product image and give it a blue background, to arrange this card's layout, and to double-check the calculation setup; all label values, calculations, references and final answers were independently verified by the student.